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Incentive Structures Theory of Everything

ISTOE: a universal root cause analysis engine.

Human behavior, organizational outcomes, and societal trends are driven overwhelmingly by upstream incentive structures and rule-sets — not by individual willpower, personal virtue, or moral failing. This edition states the framework in full technical form: its diagnostic pillars, its method, its terminal vocabulary, its known failure modes, and the limits it concedes up front.

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Diagnostic rule: Do not ask “What is wrong with the people?” Ask “What behavior is this rule-set mathematically rewarding?”

> *“Systems dictate behavior far more reliably than character.”*

Executive summary

ISTOE generalizes root-cause analysis, dependency mapping, and telemetry troubleshooting into a scale-invariant diagnostic framework — a semantic index of incentives from micro to macro. Its diagnostic rule: map every stakeholder without truncation, trace each incentive to its terminal goal, and ask what behavior the rule-set mathematically rewards, rather than pathologizing the agents operating under it.

It is a method, not a model: a method specifies how to interrogate any system whose output is in question. Its test is therefore usefulness, not completeness — does the map locate a rule that can be changed, and does changing it move the output?

What “everything” means in v1.0

The scope is stated precisely, because the name invites overreading. “Everything” means *at every scale* — one person, one team, one company, one market, one country — not every phenomenon explained. Five stated concessions (Section 7) bound what the framework claims. Absorbing them demotes a Theory of Everything to a strong, falsifiable programme — a promotion, not a defeat: a programme that can fail a test can also pass one.

The argument in four steps

1. **Diagnose upstream.** Trace behavior to the incentive structure and rule-set that makes it locally rational.
2. **Map without truncation.** Map every stakeholder; trace each incentive to its terminal goal, in the shared terminal currency of Section 5.
3. **Redesign the game.** Treat the system itself as a design variable and align stakeholder incentives so cooperation is the default equilibrium.
4. **Score by the floor.** Judge a system by its baseline — the minimum quality of life, lifespan, and healthspan it delivers — not by its average.

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1. Behavior tracks rule-sets, and 3D thinking reads them

Human behavior, organizational outcomes, and societal trends are driven overwhelmingly by upstream incentive structures and rule-sets, not by individual willpower, personal virtue, or moral failing. When the same behavior appears at scale, across different people rotated through the same position, the explanation is structural. The people changed. The behavior did not, because the rule-set did not.

> **Diagnostic rule of thumb:** Whenever analyzing a social, economic, or behavioral issue, stop asking *"What is wrong with the people?"* and instead audit: *"What behavior is this specific rule-set mathematically rewarding?"*

2D thinking flattens; 3D thinking holds the axes apart

2D thinking is binary, flat thinking: a system is good or bad, works or does not work, is a scam or a revolution — one label, one axis. **3D thinking** holds a system on multiple axes at once — incentives, capacities, and nuance — and holds divergent, non-contradictory truths in suspension. An investment trend can be a bubble on the capital axis and genuinely early on the capability axis at the same time, because those are different axes, not competing verdicts.

Most people already think in 3D somewhere: a mechanic with engines, a nurse with patients, a teacher with a classroom — incentives, capacities, and nuance, scoped to one domain. That is *Local 3D*. *General 3D* is rarer: carrying the same multi-axis lens across disciplines, in one unified syntax, without losing the local detail. ISTOE is the instrument for General 3D. It does not grant the reader a new faculty; it generalizes a faculty the reader already has locally, and gives it a transferable procedure.

The Logician Empath stance

The framework runs on two channels at once. The logician channel reads structure: dependencies, reward functions, constraint budgets, failure modes. The empath channel reads what the structure costs the agents inside it — which loops are fed, which are starved, and where the penalty actually lands. Either channel alone truncates the map. Logic without the empath channel optimizes an abstraction and misses the terminal currency; empathy without the logic channel feels the output accurately and misattributes its cause to character. The method in Section 4 is the two channels run in sequence: feel the output, then audit the architecture that produced it.

2. The four diagnostic pillars

Pillar 1 — Energy-return ratio: when the return on effort collapses, disengagement is rational

Human action is an investment of finite biological currency: time, attention, and executive bandwidth. When inputs reliably scale to outputs — for example, when labor leads to stability — agents engage and build intrinsic motivation. When the energy-return ratio collapses, through high entry costs and low or erratic payouts, disengagement is a rational energy-conservation response, not a personal pathology. “Lying flat,” NEETdom, and burnout belong in this category: they are what a correctly functioning conservation mechanism looks like when the ratio it monitors goes negative.

Pillar 2 — Upstream architecture vs. downstream blame

Mainstream discourse and media pathologize individuals to protect status-quo systems. Treating youth disengagement as a “lazy generation” problem is downstream blame. Analyzing housing costs and credential inflation is upstream architecture. The first framing protects the system by locating the fault in the one component that cannot be redesigned — a generation’s character. The second examines components that can.

Pillar 3 — Asymmetric penalties and mismatched interfaces

When individual cognitive profiles — jagged skill sets whose peaks and troughs do not match a standard template — collide with rigid, single-default systems, the interface becomes punitive. People are penalized not for a lack of capability, but for failing to navigate arbitrary compliance walls. The penalty is asymmetric: the full cost of the mismatch falls on the agent whose profile differs from the default, while the system that specified the interface bears none of it. A navigation failure is misread, systematically, as a capability failure.

Pillar 4 — Macro-economic game theory: corporate behavior follows the reward function written into law

Corporate behavior and wealth consolidation are direct responses to legal and tax rule changes. The canonical case is SEC Rule 10b-18, adopted November 17, 1982, which gave companies a safe harbor for buying back their own stock; buybacks at scale followed the rule change, not a change in corporate character. Preferential capital-gains rates over labor rates write a second reward function into the same topography. Executives are rational actors optimizing for the reward function of the system. Condemning “corporate greed” without altering the underlying topography of the rules does not change the optimization, and therefore does not change the behavior.

> The four pillars are one observation at four altitudes: agents respond to the ratio, the interface, and the reward function in front of them — and discourse blaming the agents is itself an output of the system it protects.

3. One toolkit, every domain: the four-lens template

ISTOE is not a foreign apparatus glued onto the mind. It takes the root-cause analysis, dependency mapping, and telemetry troubleshooting already used on IT networks, game engines, and digital infrastructure, and applies that same procedure as a master key — externally, to organizations and economies, and internally, to a single agent's own architecture. That root-cause procedure, generalized, is the engine the tagline names.

Run this first pass on any system, before the deeper mapping of Section 4:

Lens	The question it asks	What it maps to in ISTOE
Inputs	What drives the behavior or the error? Which pressures and collisions feed this system?	Stakeholder incentives and terminal goals — for example, high-potency core values colliding inside one agent.
Architecture	What are the pathways and delivery mechanisms? Where does the signal actually travel?	Rule-sets, interfaces, and loops — for example, an outdated approval process acting as the old pipeline every request must traverse.
Constraints	What are the hardware limits the system runs under?	The energy-return ratio and finite currency — time, attention, and energy budgets that no redesign can legislate away.
Output	What is the final system state?	Downstream symptoms — burnout, churn, disengagement, consolidation. The output is where diagnosis starts and the last place blame belongs.

A unified syntax

Because the four lenses are domain-neutral, one logical toolkit debugs a script, analyzes a corporate structure, or maps an emotional architecture, with no change of grammar between the three. A dependency in a network, a dependency in an org chart, and a dependency between two reward loops are the same shape of object: an edge in a map, with a direction and a load. That is what turns reality from a chaotic black box into a transparent, editable schematic — not a metaphor for understanding, but a literal change in what operations are available. A black box can only be complained about. A schematic can be redrawn.

4. The method: steps 0–7

Step 0 — Moral impotence. Before mapping, discount the moral framing to zero. Moral norms are social norms assigned higher priority, and they are downstream of incentive structures and determined by them — which is why right and wrong vary with geolocation, laws, and customs. The framework does not merely accommodate that diversity; it predicts it. Step zero is a debiasing procedure, not a moral claim: it divorces actions from their emotional intensity so the mechanism can be examined logically, and it is the operational refusal of Pillar 2's downstream blame.

Step 1 — Name the symptom (Output). State what is actually happening, in observable terms, with no character vocabulary in the sentence. "Completion rate fell 30% in two quarters" is a symptom. "People stopped caring" is a verdict wearing a symptom's clothes.

Step 2 — Map every stakeholder, without truncation. Include the quiet stakeholders, the rule-writers, and the system's maintainers — not only the agents being blamed. Truncation here invalidates everything downstream.

Step 3 — Trace each incentive to its terminal goal. Do not stop at instrumental goals. "Extract rent" and "maximize margin" are waypoints; what the capital is *for* — goods, services, security, loop satisfaction — is the terminal node, expressed in the vocabulary of Section 5.

Step 4 — Identify the pattern. Match the mapped structure against the pattern index of Section 6: reward hacking, energy-return collapse, crowding-out, continuity bias, mismatched interface, truncation. Naming the pattern imports everything already known about that pattern's behavior.

Step 5 — Check the known failure modes. Run the map against the failure-mode index of Section 7 before trusting it — in particular Goodhart effects on any metric the redesign will introduce, and preference falsification in any stakeholder's stated goals.

Step 6 — Treat the system as a design variable. Redesign the rule-set so the desired behavior is the default equilibrium — the locally rational choice for each agent, requiring no heroism and no sustained willpower to maintain.

Step 7 — Judge by the floor test. A redesign passes only if it raises the baseline for the worst-off stakeholder: quality of life, lifespan, and healthspan at the minimum, not the mean. If the floor did not move, the redesign did not work, whatever the averages say.

5. Every map bottoms out in the same terminal currency

Alignment is only a concrete operation if stakeholder goals can be compared in a shared unit. The framework's terminal vocabulary supplies one: every incentive, traced far enough, bottoms out in **emotional valence × nested reward loops × self-determination**.

Pleasure runs on nested loops and different timescales

All behavior is filtered through the reward center of the brain, in nested game loops across timescales: a primary loop minute to minute, a secondary loop hour to hour, a tertiary loop day to day, and longer loops beyond those. A pleasant conversation, the contentment after accomplishing something difficult, and the satisfaction of sexual desire are all the same currency running on different loops — which is precisely why loops can conflict inside one agent, a problem Section 9 carries as open.

Self-determination names the loops that matter most

Three loops carry particular weight: **autonomy**, **competence**, and **relatedness**. A worker's incentive was never just wages; it includes the autonomy and competence loops — which is why a raise that destroys autonomy can still fail to align that worker. Status and power also resolve into loop activation under this vocabulary, which changes the design question from "how do we abolish status?" to "how do we deliver that activation without rivalrous mechanics?"

Terminal element	What it contributes to a map
Emotional valence	The sign of an outcome for an agent: approach or avoid.
Nested loops	The timescale of the reward: minute, hour, day, and longer.
Autonomy / competence / relatedness	The named goods a rule-set must actually deliver, not merely signal.

Two corrections the v1.0 currency carries

First, **wanting is not liking**. The drive to pursue a reward and the pleasure of receiving it are dissociable systems (Berridge & Robinson); in addiction, wanting escalates while liking stays flat. Any map that treats pursuit as evidence of satisfaction will certify exploitative systems as aligned. Second, hedonic and eudaimonic motivation are likewise dissociable: a unitary "pleasure" fails as a terminal unit. The currency above is therefore a vocabulary, not a scalar — maps are compared loop by loop, not collapsed into a single number that a redesign could then Goodhart.

6. Three patterns recur at every scale

Underneath the four pillars sit three patterns that anchor the v1.0 semantic index. Each was first visible in one domain and then recognized, unchanged in structure, in the others — which is the root cause analysis engine's basic operation, and the evidence that the index is an index of real recurrences rather than a list of analogies.

Reward hacking

Any optimizing agent finds the shortest path between the reward signal and the reward, whether or not that path passes through the goal the signal was meant to proxy. Reward hacking is universal and evolutionary, not an artifact of machine learning: maximum reward for minimum effort is a subsistence legacy, and engineered stimuli in an environment of abundance hijack the survival loop that legacy built. The index distinguishes *self-hacking* — an agent gaming its own loops, as in the midnight scroll chosen over sleep — from *other-hacking*, where one stakeholder designs the proxy another will chase. It also tracks *proxy cascades*: chains in which each layer's metric becomes the next layer's target, so the terminal goal attenuates to nothing while every intermediate dashboard stays green.

Maximum output, minimum input

Human nature, read through the energy-return ratio, is an optimization pressure rather than a moral fact: agents route effort toward the highest return per unit of finite currency and away from collapsed ratios. Read morally, this pressure produces the downstream vocabulary of laziness and greed. Read structurally, it is the same conservation law running in every agent in the map — including the mapper, including the critic. A redesign that requires agents to override this pressure continuously is a redesign that has scheduled its own failure.

Continuity bias

Systems exhibit a strong bias toward their own continuity over adaptation or change. Continuity bias is the mechanism underneath three phenomena usually treated separately: the *transition problem* (incumbents optimized for the old game spend its rewards blocking the new one), *norm inertia* (a norm persists because deviation is priced, not because the norm is endorsed), and *policy-feedback lock-in* (a rule-set manufactures the constituency that then defends it). Any redesign proposal that does not budget for continuity bias has modeled the destination and omitted the terrain between here and there.

> Run together, the constellation explains the framework's central recurring observation: intelligent, well-intentioned agents, conserving finite energy, chasing the rewards actually on offer, and defending the arrangement they already know — inside rules somebody else set, producing an output nobody chose.

7. The framework is indexed against its own failure modes

A diagnostic method that cannot enumerate its own failure modes is asking for trust it has not earned. The v1.0 index includes, at minimum:

- **Reward hacking and proxy cascades** — the metric is satisfied, the goal is not, at one layer or through a chain of them.
- **Goodhart / Campbell effects** — a measure becomes a target and ceases to measure; the redesign's own success metric is the first candidate.
- **Crowding-out** — incentives change the preferences they act on. In Gneezy & Rustichini's daycare study, fining late pickups made them *more* common — the fine converted a social norm into a priced service — and lateness stayed high after the fine was removed.
- **Preference falsification** — stated goals diverge from revealed ones under social pressure, so a map drawn from statements alone is a map of the pressure.
- **Within-agent loop conflicts** — short and long loops in one agent want opposite things; between-agent alignment does not settle them.
- **Stakeholder-map truncation** — stopping at instrumental goals, or treating the current system as fixed background. The method's named, primary failure mode.
- **The transition problem** — the path to a redesigned equilibrium must survive agents optimized for the current one.

The five concessions

1. Motivation is plural and dissociable; a unitary pleasure theory fails as stated.
2. Incentives change the preferences, norms, and identity they act on, and the change is sometimes irreversible.
3. Perfect mapping and alignment are impossible by theorem — Arrow, Gibbard–Satterthwaite, and Myerson–Satterthwaite bound what any alignment procedure can achieve.
4. No neutral mapper exists; every mapping is positioned, including this framework's.
5. "Raise the floor" and "genuine satisfaction" are contested moral choices, stated openly — and their measurement is Goodhart-vulnerable.

8. Align all stakeholders so cooperation is the locally rational choice

The prescriptive goal is not a party or a policy platform. It is aligning the incentive structures of **all** involved stakeholders to change the default behavior of the system, so the desired behavior becomes the locally rational choice for each agent. Solutions are not parties or platforms; alignment is nonpartisan by construction, because parties are actors optimizing inside rule-sets, not the rule-sets themselves.

Apparent zero-sum conflicts are artifacts of truncated maps

Landlord and tenant on rent, or employer and worker on wages, look irreducibly zero-sum only under truncation. Capital-extractive behavior is still motivated by acquiring capital to spend on goods and services. Trace that incentive to its terminal goal, and include the system itself as a design variable, and the zero-sum appearance dissolves. A conflict may only be declared irreducible after full mapping — and, per concession 3, some will survive that test. The method's claim is narrower and stronger than "every conflict dissolves": most declared-irreducible conflicts have never been fully mapped.

A rule-set must reward loop satisfaction, not loop exploitation

Current systems already reward-hack human loops. Engagement algorithms, gambling mechanics, and doomscrolling trigger reward loops without delivering the terminal goods: the hit without the autonomy, the notification without the relatedness — wanting activated, liking and satisfaction withheld. A prescriptive pillar that only said "satisfy the loops" could bless that exploitation, so the design criterion is explicit: reward genuine loop *satisfaction*, not loop *activation* alone. The framework never needs to call exploitation evil. Exploitation is a structural misalignment, and the prescription is redesign, not condemnation.

Score a system by its floor, not its average

The design philosophy of a successful implementation is to raise the floor: maximize the baseline of human existence across **quality of life, lifespan, and healthspan**. A system's score is its minimum, not its mean. This does three things. It resolves the tension between maximizing an average and aligning all stakeholders — a pure average can license immiserating a minority to raise the mean, and a floor cannot. It binds the mapper: whoever draws the map has their own reward loops, so the process is constrained by an explicit, checkable objective rather than by trust in a neutral agent. And the trio is wireheading-resistant by construction: pure pleasure-maximization could endorse stimulating the reward center while bodies rot, but lifespan and healthspan rule that out structurally.

> **The spine of the theory:** the diagnostic half is the story of floor collapse — Pillar 1's energy-return ratio describes a baseline of existence becoming unlivable, with disengagement as the rational response. The prescriptive half is the mirror image: floor-raising.

9. Open questions the theory has not yet closed

The concessions of Section 7 bound what the framework claims. The questions below are different: they are places where the framework's own machinery generates a problem it has not yet solved. They are carried openly, as the refinement agenda.

- **Mapper reflexivity:** whoever draws the map has their own reward loops. Participatory mapping — stakeholders mapping themselves, with the mapper's incentives audited in the same pass — is a candidate mitigation, not yet settled.
- **Strategic misreporting:** once redesign follows mapping, stakeholders can game their reported terminal goals. Mechanism design's strategy-proofness results should be imported rather than re-derived.
- **Within-agent loop conflicts:** the loop theory creates this one itself. Short and long loops in one person can want opposite things, as in addiction. The method adjudicates *between* agents; it needs an adjudication rule *within* them.
- **The transition problem:** incumbents optimized for the old game can use its rewards to block redesign. The theory describes a destination; it needs a path that survives the old equilibrium — continuity bias, Section 6, is the terrain that path must cross.
- **Operationalizing "genuine":** satisfaction versus exploitation needs operational teeth, or the criterion risks becoming moral language in structural dress — which would violate step zero from inside the framework.
- **AI-alignment generality:** agents reward-hacking because top-of-stack incentives propagate down into the model is this same framework applied to machines. If the index holds there too, it is a candidate generality proof for the engine itself.

The one-paragraph version

> Strip the moral charge. Name the output. Map every stakeholder without truncation and trace each incentive to its terminal goal, in the currency of valence, nested loops, and self-determination. Name the pattern, check the failure modes, and redesign the rule-set so the desired behavior is the default — then judge the redesign by whether it raised the floor. That procedure, run at any scale, in one unified syntax, is ISTOE: a universal root cause analysis engine.

Version 1.01 — public edition, November 2026. This edition supersedes the v1.0 public edition and the v0.1 white paper (October 2026): the scope is restated as a diagnostic framework and semantic index; the four-lens template, the pattern constellation, the known failure modes, and the five concessions are incorporated; the tagline is updated to "a universal root cause analysis engine." Diagnostic pillars, the prescriptive method, and the floor objective reflect positions confirmed by the author. Section 9 lists questions the author has left open for further refinement.